Lecture 10. Monte Carlo Integration

Naive Monte Carlo, importance sampling, antithetic variables, control variables

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10.1 Naive Monte Carlo

To compute $\mu = \mathbb{E}f(X) = \int f(x)p(x)\,dx$ for a density p, draw $X_1, \dots, X_n$ i.i.d. from p and use

$$\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} f(X_i).$$

It is unbiased; by the strong law $\hat{\mu}_n \to \mu$ almost surely when $\mathbb{E}|f(X)| < \infty$; and if $\sigma^2 = \operatorname{Var} f(X) < \infty$, the CLT gives $\sqrt{n}(\hat{\mu}_n - \mu) \Rightarrow N(0, \sigma^2)$. With the sample variance $\hat{\sigma}^2$ of the $f(X_i)$, an approximate 95% interval is $\hat{\mu}_n \pm 1.96\,\hat{\sigma}/\sqrt{n}$.

The error is $O_p(n^{-1/2})$ *regardless of the dimension*, while deterministic quadrature on a grid suffers from the curse of dimensionality. In one or two dimensions, quadrature (Lecture 8) is far more accurate; Monte Carlo wins in high dimensions. To improve on $n^{-1/2}$ one cannot do much, but the constant σ^2 can be reduced: this is **variance reduction**.

Example 10.1

- $\int_0^\infty \sqrt{x}e^{-x}dx = \Gamma(3/2) = \sqrt{\pi}/2$ is $\mathbb{E}\sqrt{X}$ for $X \sim \text{Exp}(1)$; the average of $\sqrt{X_i}$ estimates it with finite variance.
- $\int_0^1 x^{-1/2}dx = 2$ is $\mathbb{E}U^{-1/2}$, but $\mathbb{E}U^{-1} = \infty$: the estimator is consistent yet has infinite variance, so the CLT interval is meaningless and the running averages show occasional large jumps. The substitution $x = y^2$ turns the integral into $\int_0^1 2\,dy$, estimated exactly. A transformation that flattens the integrand is the cheapest variance reduction.
- Buffon's needle: a needle of length L dropped on a floor ruled with parallel lines at spacing $D \geq L$ crosses a line with probability $2L/(\pi D)$. The centre's distance to the nearest line is $Y \sim \text{Unif}(0, D/2)$ and the angle $\Theta \sim \text{Unif}(0, \pi/2)$, independent; the needle crosses iff $Y \leq \frac{L}{2}\sin\Theta$, which has probability $\frac{2}{D} \cdot \frac{2}{\pi}\int_0^{\pi/2} \frac{L}{2}\sin\theta\,d\theta = \frac{2L}{\pi D}$. The crossing frequency thus estimates π: with k crossings in n throws, $\hat{p} = k/n$ is unbiased for $p = 2L/(\pi D)$ and $\hat{\pi} = 2L/(D\hat{p}) = 2Ln/(Dk)$. But $\hat{\pi}$ is *not* unbiased: $1/x$ is convex, so $E[1/\hat{p}] > 1/p$ by Jensen, and in fact $E\hat{\pi} = \infty$ because $k = 0$ has positive probability. By the delta method with $g(p) = 2L/(Dp)$, $g'(p)^2 = \{2L/(Dp^2)\}^2 = \pi^2/p^2$, so

$$\operatorname{Var}\hat{\pi} \approx g'(p)^2 \frac{p(1-p)}{n} = \frac{\pi^2(1-p)}{np},$$

which decreases in p; the best choice is $L = D$, giving $\operatorname{Var}\hat{\pi} \approx 5.63/n$. Even then, three correct digits need tens of millions of throws.

10.2 Importance sampling

For any density h with $h > 0$ wherever $fp \neq 0$,

$$\mu = \int f(x)\frac{p(x)}{h(x)}h(x)\,dx = \mathbb{E}_h\Big[f(Y)\frac{p(Y)}{h(Y)}\Big], \qquad \hat{\mu}_h = \frac{1}{n}\sum_{i=1}^{n} f(Y_i)w(Y_i), \quad Y_i \overset{\text{iid}}{\sim} h,$$

with **importance weights** $w = p/h$. It is unbiased with variance $\frac{1}{n}\{\int f^2p^2/h - \mu^2\}$. This is useful when p is hard to sample, or when p rarely visits the region where f is large.

Proposition 10.1

The variance of $\hat{\mu}_h$ is minimized by $h^\star \propto |f|\,p$. If $f \geq 0$, then $h^\star = fp/\mu$ and the variance is zero.

Proof. By Cauchy–Schwarz, for every admissible h,

$$\Big(\int |f|p\Big)^2 = \Big(\int \frac{|f|p}{\sqrt{h}}\sqrt{h}\Big)^2 \leq \int \frac{f^2p^2}{h}\int h = \int \frac{f^2p^2}{h},$$

with equality iff $|f|p/\sqrt{h} \propto \sqrt{h}$, i.e. $h \propto |f|p$. If $f \geq 0$, $fw \equiv \mu$ under $h^\star$. ■

$h^\star$ needs μ itself, so it is unusable, but it is a guide: choose h roughly proportional to $|f|p$, and with tails at least as heavy as fp; otherwise the weights explode and the variance may be infinite. A useful diagnostic is the spread of the weights.

Example 10.2 (Normal tail probability)

Estimate $\pi(c) = P(Z > c) = \Phi(-c)$ for large c. The naive estimator $\frac{1}{n}\sum \mathbf{1}\{Z_i > c\}$ has relative error $\sqrt{(1-\pi(c))/(n\pi(c))}$, which explodes as c grows: almost no draw lands in the tail. Shift the sampling distribution to $h = \phi(\cdot - c)$, i.e. $Y_i \sim N(c,1)$. The weight is $\phi(y)/\phi(y-c) = e^{-cy+c^2/2}$, so

$$\hat{\pi}_{\text{imp}}(c) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{Y_i > c\}e^{-cY_i+c^2/2}.$$

Its second moment per draw is

$$e^{c^2}\int_c^\infty e^{-2cy}\phi(y-c)\,dy = e^{c^2}\int_c^\infty \phi(y+c)\,dy = e^{c^2}\Phi(-2c),$$

since $-2cy - \frac{1}{2}(y-c)^2 = -\frac{1}{2}(y+c)^2$. Hence $\operatorname{Var}\hat{\pi}_{\text{imp}}(c) = \frac{1}{n}\{e^{c^2}\Phi(-2c) - \pi(c)^2\}$. By Mills' ratio $\Phi(-t) \approx \phi(t)/t$ as $t \to \infty$, the squared relative error is about $c\sqrt{\pi/2}/n$, against $\sqrt{2\pi}\,ce^{c^2/2}/n$ for the naive estimator: the relative error grows like $c^{1/2}$ instead of $c^{1/2}e^{c^2/4}$.

10.3 Antithetic variates

If X and X' have the same distribution but are negatively correlated, then

$$\operatorname{Var}\Big[\frac{f(X)+f(X')}{2}\Big] = \frac{\operatorname{Var} f(X)}{2}(1+\rho), \qquad \rho = \operatorname{Corr}(f(X), f(X')),$$

which beats two independent draws when $\rho < 0$. For uniforms the natural pair is $(U, 1-U)$; for symmetric distributions about m, $(X, 2m-X)$.

Proposition 10.2

If f and g are both nondecreasing (or both nonincreasing), then $\operatorname{Cov}(f(X), g(X)) \geq 0$ for any random variable X for which the covariance exists. Consequently, for monotone h, $\operatorname{Cov}(h(U), h(1-U)) \leq 0$.

Proof. Let Y be an independent copy of X. Since f and g move in the same direction, $\{f(X) - f(Y)\}\{g(X) - g(Y)\} \geq 0$. Its expectation is

$$2\mathbb{E}f(X)g(X) - 2\mathbb{E}f(X)\mathbb{E}g(X) = 2\operatorname{Cov}(f(X), g(X)) \geq 0.$$

For the consequence, take $f = h$ and $g(u) = -h(1-u)$, which is monotone in the same direction as h. ■

Example 10.3

For $\mu = \mathbb{E}e^U = e - 1$: $\operatorname{Var} e^U = \frac{e^2-1}{2} - (e-1)^2 \approx 0.2420$ and $\operatorname{Cov}(e^U, e^{1-U}) = e - (e-1)^2 \approx -0.2342$. The antithetic average $\frac{1}{2}(e^U + e^{1-U})$ has variance $\frac{1}{2}(0.2420 - 0.2342) \approx 0.0039$, against $0.2420/2 = 0.1210$ for two independent uniforms: a reduction of about 97%.

10.4 Control variates

Suppose $Y = g(X)$ has a known mean μ_Y. For every c, $f(X) - c(Y - \mu_Y)$ is unbiased for μ, with variance

$$\operatorname{Var} f(X) + c^2 \operatorname{Var} Y - 2c \operatorname{Cov}(f(X), Y),$$

minimized at $c^\star = \operatorname{Cov}(f(X), Y)/\operatorname{Var} Y$, where it equals $\operatorname{Var} f(X)\,(1 - \rho^2)$, $\rho = \operatorname{Corr}(f(X), Y)$. The more correlated the control, the larger the gain. In practice $c^\star$ is estimated by the least-squares slope of $f(X_i)$ on Y_i (a small bias, negligible for large n).

Example 10.4

For $\mathbb{E}e^U$ with control $Y = U$ ($\mu_Y = \frac{1}{2}$, $\operatorname{Var} U = \frac{1}{12}$): $\operatorname{Cov}(e^U, U) = \mathbb{E}Ue^U - \frac{e-1}{2} = 1 - \frac{e-1}{2} \approx 0.1409$, so $c^\star = 12 \times 0.1409 \approx 1.6903$, and the variance per draw drops from 0.2420 to $0.2420 - 12(0.1409)^2 \approx 0.0039$.

10.5 A comparison: the size of a test

Example 10.5

With $X_1, \ldots, X_{25}$ i.i.d. Poisson(λ), the CLT test of $H_0 : \lambda = 2$ against $\lambda > 2$ rejects when $Z = (\bar{X} - 2)/\sqrt{2/25} > 1.645$, i.e. when $S = \sum X_i > c = \lfloor 25(2 + 1.645\sqrt{2/25}) \rfloor = 61$. Under λ, $S \sim \text{Poi}(25\lambda)$, so the size is $p = 1 - F_{50}(61) = 0.05568$, slightly above the nominal 0.05. Pretend we estimate p by simulation with N draws of S.

- *Standard*: $\hat{p} = \frac{1}{N}\sum \mathbf{1}\{S_j > c\}$ has variance $p(1-p)/N = 0.0526/N$.
- *Antithetic*: with $S_j = F_{50}^{-1}(U_j)$ and $S_j' = F_{50}^{-1}(1 - U_j)$, the two indicators cannot both be 1 (that would need $F_{50}(c) < U_j < 1 - F_{50}(c)$, impossible since $F_{50}(c) > \frac{1}{2}$), so their covariance is $-p^2$, and the average of N pairs has variance $p(1-2p)/(2N) = 0.0247/N$, compared with $p(1-p)/(2N) = 0.0263/N$ for $2N$ independent draws. That is only a 6% gain: the indicator of a rare event is a monotone function but nearly constant, so the negative correlation is weak.
- *Importance sampling*: draw $S_j \sim h = \text{Poi}(25\lambda')$ with $\lambda' = 2.4653$, the rejection threshold, so that about half the draws land in the rejection region. The weight is $w(s) = e^{-25(\lambda - \lambda')}(\lambda/\lambda')^s$, and the variance per draw is $\sum_{s>c} f_{50}(s)^2/h(s) - p^2 = 0.0063$, with f_{50} the Poi(50) pmf: 8.3 times smaller than standard Monte Carlo.

Each method gives a CLT confidence interval $\hat{p} \pm z_{1-\alpha/2}\sqrt{\widehat{\text{Var}}}$ with the variance estimated from the simulated summands. For the power at $\lambda \in [2.2, 4]$, the same code applies with F_{50} replaced by $F_{25\lambda}$. Once the rejection region is no longer rare ($\lambda \geq 2.5$, power ≥ 0.5), the advantage of the fixed envelope $\lambda' = 2.4653$ disappears. For large λ the weights $(\lambda/\lambda')^s$

become extremely variable and importance sampling becomes *worse* than standard Monte Carlo, whose variance $p(1-p)/N$ shrinks as the power approaches 1. Per draw, the importance sampling variance is 0.299 against 0.248 at $\lambda = 2.5$, and 17.3 against 0.053 at $\lambda = 3$. An envelope tuned to one λ is no good for all λ.

10.6 Selected exercises

Motivation. Stratification is a further variance reduction method; self-normalized weights make importance sampling usable when the target density is known only up to a constant; and a Gaussian example shows exactly how too-light tails of the sampling density ruin the variance.

Exercise 10.1 (Stratified sampling)

Partition the sample space into strata $A_1, \dots, A_J$ with known $p_j = P(X \in A_j) > 0$. Let $\mu_j = \mathbb{E}[f(X) \mid X \in A_j]$ and $\sigma_j^2 = \mathrm{Var}[f(X) \mid X \in A_j]$. Draw $n_j = np_j$ (assumed integer) independent points from the conditional distribution of X given $X \in A_j$, independently across strata, let $\bar{f}_j$ be the average of f over them, and put $\hat{\mu}_{\mathrm{st}} = \sum_j p_j \bar{f}_j$.

(a) Show that $\hat{\mu}_{\mathrm{st}}$ is unbiased with $\mathrm{Var}\,\hat{\mu}_{\mathrm{st}} = \frac{1}{n}\sum_j p_j \sigma_j^2 \le \frac{\sigma^2}{n}$, with equality iff all μ_j are equal.

(b) For $X \sim \mathrm{Unif}(0,1)$, strata $[\frac{j-1}{J}, \frac{j}{J})$ and $|f'| \le K$, show $\mathrm{Var}\,\hat{\mu}_{\mathrm{st}} \le K^2/(4nJ^2)$.

Solution. [Supplement]

(a) $\mathbb{E}\bar{f}_j = \mu_j$, so $\mathbb{E}\hat{\mu}_{\mathrm{st}} = \sum_j p_j \mu_j = \mathbb{E}f(X) = \mu$ by the law of total expectation. The $\bar{f}_j$ are independent with $\mathrm{Var}\,\bar{f}_j = \sigma_j^2/n_j$, so

$$\mathrm{Var}\,\hat{\mu}_{\mathrm{st}} = \sum_j p_j^2 \frac{\sigma_j^2}{np_j} = \frac{1}{n}\sum_j p_j \sigma_j^2.$$

By the law of total variance, $\sigma^2 = \mathrm{Var}\,f(X) = \sum_j p_j \sigma_j^2 + \sum_j p_j(\mu_j - \mu)^2 \ge \sum_j p_j \sigma_j^2$, with equality iff $\mu_j = \mu$ for all j. Stratification removes the between-strata part of the variance.

(b) On a stratum of length $1/J$, f takes values in an interval of length at most K/J by the mean value theorem. A random variable with values in an interval of length ℓ has variance at most $(\ell/2)^2$ (its mean squared deviation from the interval's midpoint is at most $(\ell/2)^2$). So $\sigma_j^2 \le K^2/(4J^2)$ and, by (a), $\mathrm{Var}\,\hat{\mu}_{\mathrm{st}} \le K^2/(4nJ^2)$. With $J = n$ (one point per stratum) the variance is $O(n^{-3})$, against $O(n^{-1})$ for naive Monte Carlo. ■

Exercise 10.2 (Self-normalized importance sampling)

Suppose $p = \tilde{p}/Z$ with $\tilde{p}$ computable but Z unknown, and $h > 0$ wherever $p > 0$. Put $\tilde{w} = \tilde{p}/h$, $w = p/h$, and with $Y_1, \dots, Y_n \overset{\mathrm{iid}}{\sim} h$,

$$\tilde{\mu}_n = \frac{\sum_i f(Y_i)\tilde{w}(Y_i)}{\sum_i \tilde{w}(Y_i)}.$$

(a) Show that $\tilde{\mu}_n \to \mu = \mathbb{E}_p f(X)$ almost surely when $\mathbb{E}_p|f| < \infty$.

(b) If $\mathbb{E}_h[w^2(f-\mu)^2] < \infty$, show that $\sqrt{n}(\tilde{\mu}_n - \mu) \Rightarrow N(0, \mathbb{E}_h[w(Y)^2\{f(Y)-\mu\}^2])$.

Solution. [Supplement]

(a) $\mathbb{E}_h[f\tilde{w}] = Z\int fp = Z\mu$ and $\mathbb{E}_h[\tilde{w}] = Z$, both finite. By the strong law, the numerator and denominator divided by n converge almost surely to $Z\mu$ and $Z > 0$, so their ratio converges to μ. The unknown Z cancels in $\tilde{\mu}_n$.

(b) Write

$$\sqrt{n}(\tilde{\mu}_n - \mu) = \frac{n^{-1/2}\sum_i \tilde{w}(Y_i)\{f(Y_i) - \mu\}}{n^{-1}\sum_i \tilde{w}(Y_i)}.$$

The summands in the numerator are i.i.d. with mean $Z(\mu - \mu) = 0$ and variance $Z^2\mathbb{E}_h[w^2(f - \mu)^2] < \infty$, so by the CLT the numerator converges in distribution to $N(0, Z^2\mathbb{E}_h[w^2(f-\mu)^2])$. The denominator converges to Z almost surely, and Slutsky's theorem gives the claim. The asymptotic variance can be estimated by $\sum_i \tilde{w}_i^2(f_i - \tilde{\mu}_n)^2/(\sum_i \tilde{w}_i)^2$, i.e. with the normalized weights. ■

Exercise 10.3 (Importance densities with light tails)

Let $p = N(0,1)$ and $h = N(0,s)$ with variance $s > 0$, and $w = p/h$.
(a) Show that $\mathbb{E}_h[w(Y)^2] < \infty$ if and only if $s > \frac{1}{2}$, and compute it.
(b) Using $f \equiv 1$ (so that the estimator is the average weight, estimating 1), find the variance per draw and the best s.

Solution. [Supplement]

(a) $\mathbb{E}_h[w^2] = \int p^2/h$, and

$$\frac{p(x)^2}{h(x)} = \frac{e^{-x^2}/(2\pi)}{e^{-x^2/(2s)}/\sqrt{2\pi s}} = \sqrt{\frac{s}{2\pi}}\exp\Big\{-x^2\Big(1 - \frac{1}{2s}\Big)\Big\}.$$

If $s \le \frac{1}{2}$ the exponent is ≥ 0 and the integral diverges. If $s > \frac{1}{2}$, with $\int e^{-cx^2}dx = \sqrt{\pi/c}$ for $c = 1 - \frac{1}{2s} > 0$,

$$\mathbb{E}_h[w^2] = \sqrt{\frac{s}{2\pi}}\sqrt{\frac{\pi}{1 - 1/(2s)}} = \frac{s}{\sqrt{2s-1}}.$$

(b) The variance per draw is $\mathbb{E}_h[w^2] - 1 = s/\sqrt{2s-1} - 1$. The derivative of $s/\sqrt{2s-1}$ is $(s-1)/(2s-1)^{3/2}$, so the variance decreases on $(\frac{1}{2}, 1)$, increases on $(1, \infty)$, and is 0 at $s = 1$ ($h = p$). It grows without bound as $s \downarrow \frac{1}{2}$ and is infinite for $s \le \frac{1}{2}$, although the estimator remains unbiased and consistent: a sampling density with lighter tails than $|f|p$ produces rare, enormous weights. Heavier tails ($s > 1$) cost only a moderate factor, e.g. $\frac{2}{\sqrt{3}} - 1 \approx 0.155$ at $s = 2$. ■